

HW/Lab 4: OFDM Decoding

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Goals and Instructions

The goals for this assignment are:

- Understand how OFDM signals are decoded.
- Implement an OFDM Receiver
- Test OFDM on SDR.

Follow these instructions:

- **Collaboration:** You can discuss concepts, but write your own solutions and code.
- **Tools:** We will primarily use MATLAB (and/or Python). Directions use standard MATLAB functions. *Strongly recommended* toolboxes: Signal Processing, Communications. (Base-MATLAB fallbacks are acceptable.)
- **Hardware:** We use PlutoSDR or Pluto+ SDR for this exercise.
- **Submission:** Upload a single PDF with your answers and plots to Gradescope before the deadline. Also compress (into zip file) and upload your finished 'lab4 starter code' folder to a separate Gradescope portal before the same deadline.

Matlab Simulations: OFDM Decoding

Learning goals. In this exercise you will (i) simulate packet detection in OFDM, (ii) add channel estimation and equalization, (iii) add CFO correction, (iv) add SFO and phase error correction, and (v) Plot constellation diagram, EVM diagram, Channel estimation, and SNR vs BER, with/without CFO correction/SFO and phase error correction.

Starter code

You are given a starter MATLAB script 'hw4.m' (On Piazza) with a library folder 'lib' of functions, which simulates the OFDM transmission. The default parameter settings in 'hw4.m' are as follows.

```

1 %% Channel Flags and Parameters
2 SIM_FLAG = 1;      % flag to run simulation, 0 to run SDR
3 CFO_FLAG = 0;      % flag to add CFO in simulation
4 SFO_FLAG = 0;      % flag to add SFO in simulation
5 CHANNEL_FLAG = 0;  % flag to apply channel in simulation
6
7 fs = 1e6; % sampling rate
8 fc = 2.45e9; % carrier frequency, roughly middle of ISM band
9 CFO = 0; % CFO in Hz
10 SFO = 0; % SFO in Hz
11 SNRdB = Inf; % awgn SNR (dB) in channel; set inf for no noise
12 ofdm_params.MOD_ORDER = 2; % Modulation order (2/4/16/64 = BPSK/QPSK
    /16-QAM/64-QAM)
13 ofdm_params.N_OFDM_SYMS = 50; % Number of OFDM symbols per packet
14
15 %% Decoder Flags and Parameters
16 ofdm_params.DO_DECODE = 1;
17 ofdm_params.DO_APPLY_CFO_CORRECTION = 0;
18 ofdm_params.DO_APPLY_SFO_CORRECTION = 0;
19 ofdm_params.DO_APPLY_PHASE_ERR_CORRECTION = 0;
20 ofdm_params.plot_flag = 1; % Plot all figures
21 ofdm_params.num_packets = 10; % Number of packets to decode
22 ofdm_params.packet_step_size = 1; % keep it 1 in hw4

```

What you need to implement

There are six functions outside ‘lib’ given as ‘p’ P-code files, which are content-obscured, execute-only MATLAB codes. You need to complete the Matlab scripts (following the comments inside) in the folder ‘decoding functions’ to replace these P-code functions. Every time you complete a function, drag it out from the ‘decoding functions’ folder, and drag the corresponding P-code function into the ‘decoding functions’ folder, so the completed function script instead of the P-code can be run by ‘hw4.m’. Please follow the steps below.

1. Complete and replace functions ‘rx_pack_detect.m’ and ‘rx_demodulation.m’. Run ‘hw4.m’ with default parameter settings to ensure it generates plots equal to the P-code files.
2. Complete and replace functions ‘rx_channel_estimation.m’ and ‘rx_equalize_signal’. Test them by running ‘hw4.m’ with parameter

CHANNEL_FLAG = 1;

3. Complete and replace function ‘rx_cfo_correction.m’. Set parameters

CFO_FLAG = 1;

CFO=100;

CHANNEL_FLAG = 0;

Then test the function by toggling the CFO correction flag

ofdm_params.DO_APPLY_CFO_CORRECTION

4. (ECSE 6560 only) Complete and replace function 'rx_sfo_and_phase_error_correction.m'.
Set parameters

CFO_FLAG = 0;

SFO_FLAG = 1;

SFO=100;

Then test the function by toggling the SFO and residual phase error correction flags

ofdm_params.DO_APPLY_SFO_CORRECTION

ofdm_params.DO_APPLY_PHASE_ERROR_CORRECTION

Deliverables (plots & explanations)

Please follow the steps below.

1. Set the parameters back to the default settings, and turn on the Channel and CFO without CFO correction

CFO_FLAG = 1;

CHANNEL_FLAG = 1;

ofdm_params.DO_APPLY_CFO_CORRECTION=0;

What do you observe in the constellation diagram, EVM diagram, and channel estimate plots?
How does CFO cause these? Submit the plots with your answer.

2. Now turn on CFO correction, and turn on SFO without correction

ofdm_params.DO_APPLY_CFO_CORRECTION=1;

SFO_FLAG = 1;

What do you observe in the constellation diagram, EVM diagram, and channel estimate plots?
How does SFO cause these? Submit the plots with your answer.

3. Now turn on the SFO and phase error corrections, and increase the number of OFDM symbols per packet

ofdm_params.DO_APPLY_SFO_CORRECTION=1;

ofdm_params.DO_APPLY_PHASE_ERR_CORRECTION=1;

params.N_OFDM_SYMS = 1000;

What do you observe in the constellation diagram, EVM diagram, and channel estimate plots?
How does a larger packet size cause these? Submit the plots with your answer.

4. Now, set back to 50 OFDM symbols per packet

params.N_OFDM_SYMS = 50;

and keep other parameters the same (which means all errors and corrections are ON), plot BER under SNR(dB) = 0:2:30. Note that you need to turn off the plotting

ofdm_params.plot_flag = 0;

and generate a FOR loop to traverse the settings of the SNR parameter

SNRdB

In each iteration, you can obtain an array of the BER for each decoded packet from the variable

op_struct.berratio

Take the mean of this array to get the average BER for each SNR value. Plot BER for BPSK

ofdm_params.MOD_ORDER = 2;

and QPSK

ofdm_params.MOD_ORDER = 4;

Are their BER values similar? Submit your answer with the BER plots.

Hardware Experiment

Learning goals. In this exercise, you will test your decoding functions on an SDR. Set up your PlutoSDR similar to Lab 3. Connect it to your computer and make sure it is programmable through Matlab. We have provided the PlutoSDR code for this experiment.

Deliverables (plots & explanations)

Please follow the steps below.

1. Set the parameters back to the default settings, connect your SDR to the computer, and turn off the simulation to run the transmission on the SDR

SIM_FLAG = 0;

Turn on all corrections

ofdm_params.DO_APPLY_CFO_CORRECTION=1;

ofdm_params.DO_APPLY_SFO_CORRECTION=1;

ofdm_params.DO_APPLY_PHASE_ERR_CORRECTION=1;

What do you observe in the constellation diagram, EVM diagram, and channel estimate plots? Submit your comments with the plots.

2. Similar to the last step in MATLAB simulation, plot BER for different SDR transmitter gains in the range $-30 : 2 : 0$. You can change the SDR transmitter gain in 'sdrtx()' function. Submit the BER plot.